

Agent: Gas Leakage detection

Introduction: A Gas Leakage Detection and Automatic Shut-Off Agent is an automated safety system designed to detect gas leakage and take immediate action.

Problem Statement: Gas leakage is a common household safety problem that may go unnoticed, especially when people are not present in the kitchen or nearby.

Problem Formulation: The problem can be formulated as an environment monitoring and decision-making problem. The agent receives gas-level readings from a gas sensor and compares them with a predefined safe limit.

Problem Solution: The proposed solution is an automated rule-based agent that continuously monitors gas sensor data.

The agent follows these steps:

Gas Sensor → Read Gas Level → Apply Rule → Detect Leakage → Shut Valve + Alarm + Warning.

Decision Rules:

IF gas level > safe limit

THEN

Gas Leakage = Detected

Gas Valve = OFF

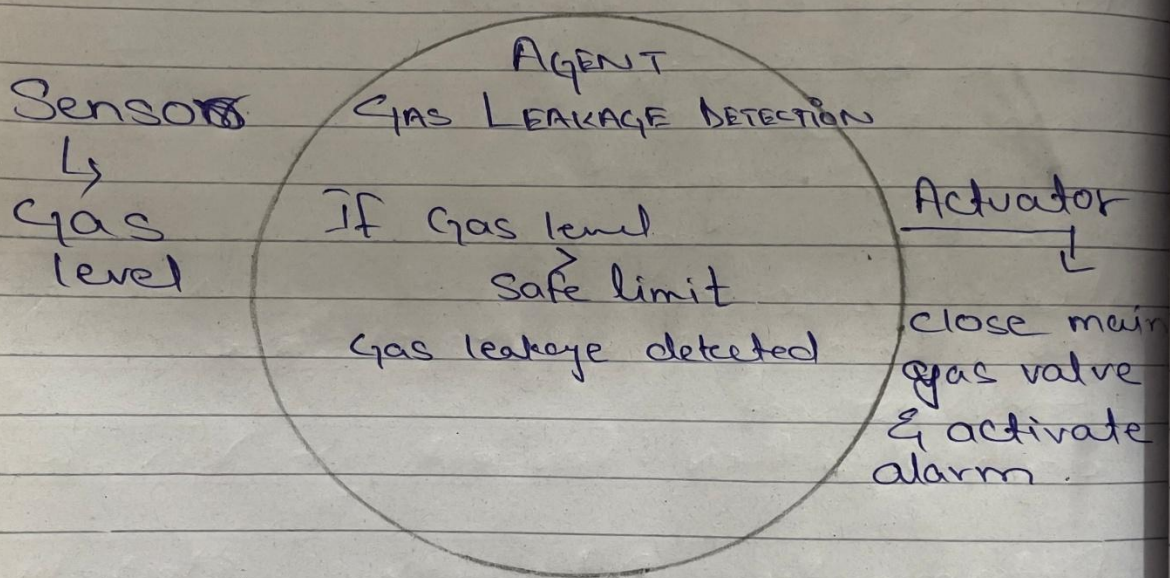
Alarm = ON

Warning Message = "Gas Leakage Detected!"

Else : Normal

Assignment

Agent : Gas leakage detection



Sensor :

- Gas sensor reading
- Temperature / Humidity
- Leakage status

Actuator :

Gas valve : OFF

Alarm : ON

PEAS Framework:

Performance: Detect leakage accurately, respond quickly

Environment: Kitchen

Actuators: Automatic gas shut-off valve

Sensors: Gas leakage sensor

Property	Type	Why
Observability	Partially Observable	The agent can't see the full real picture of the environment..
Determinism	Stochastic	Gas concentration and leakage conditions can change unpredictably.
Episodes	Episodic	Each gas detection event can be handled on its own.
Change	Dynamic	The gas level can change at any time while the agent is monitoring the environment.
Values	Continuous	Gas concentration can have many different numerical values.
Agents	Single Agent	The gas leakage detector acts as the main agent monitoring the environment.

ASSIGNMENT: RULE BASE AGENT

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COURSE: ARTIFICIAL INTELLIGENCE